

International Cybersecurity and Digital Forensics Academy (ICDFA)**CIP-B101 Basic Computer Skills for Digital Forensics****Lab 2A: Linux Command Line for Digital Forensics Basics***Student Practical Laboratory Manual*

Item	Details
Course Code	CIP-B101
Course Title	Basic Computer Skills for Digital Forensics
Lab Code	Lab 2A
Lab Title	Linux Command Line for Digital Forensics Basics
Estimated Duration	2 to 3 hours
Mode	Individual practical lab with screenshots and short written explanation
Submission Format	One PDF report plus compressed evidence workspace (.zip or .tar.gz)

Important Pre-Lab Instruction: Students must go through the provided Linux OS for Forensics slide walkthrough before starting this lab. The slide explains the Linux filesystem, VFS, inodes, paths, \$PATH, file operations, grep, networking, scripting, permissions, and compression concepts that are required for this practical work.

Forensic Safety Note: All commands must be executed only on your own Linux/Kali VM or authorized classroom lab system. Do not scan, probe, or connect to external systems except where the lab explicitly uses localhost or public reference websites for learning.

1. Lab Scenario

You are a junior digital forensic trainee assigned to prepare a Linux-based evidence workspace. Your task is to demonstrate that you can navigate the Linux filesystem, understand evidence locations, inspect files and metadata, create and manage evidence folders, search for information, check network configuration, write a simple shell script, and preserve your work in a compressed archive.

This lab focuses on Linux command-line skills that are foundational for future forensic work such as disk imaging, file recovery, evidence search, log analysis, and tool automation.

2. Learning Outcomes

- ☐ Explain common operating system functions and why Linux is useful in digital forensics.
- ☐ Identify the Linux filesystem structure and explain the idea that “everything is a file”.
- ☐ Use inode and disk-free commands to inspect file and filesystem information.
- ☐ Navigate Linux directories using absolute and relative paths.
- ☐ Create, copy, rename, delete, search, compress, and extract files using command-line tools.
- ☐ Use grep to search evidence text and document results.
- ☐ Check basic network information and test only authorized local connectivity.
- ☐ Create and execute a simple shell script, then make it callable using the PATH variable.

3. Required Lab Environment

Requirement	Description
Operating System	Kali Linux, Ubuntu, Debian, or another Linux distribution approved by the instructor.
User Access	Normal user account with sudo access for installation-only tasks.
Tools	Terminal/Terminator, tree, grep, nano or leafpad, ss/netstat, ping, nmap, wget, zip/unzip, tar.
Internet	Optional; required only for installing missing packages or visiting reference links.
Evidence Workspace	A folder created by the student under the home directory.

```
# Optional package installation if tools are missing
sudo apt update
sudo apt install -y tree nmap wget zip unzip tar net-tools terminator nano
```

4. Slide Reference URLs and Reading Materials

The following links were referenced in the slide material and may be used for further reading. Students are not required to memorize them, but they should understand the concepts before submitting the lab.

- Operating system overview: https://en.wikipedia.org/wiki/Operating_system
- File structure illustration: http://home.easy-key.info/images/stories/file_structure.gif
- IBM Virtual File System overview: <https://developer.ibm.com/technologies/linux/tutorials/l-virtual-filesystem-switch/>
- Inode explanation: <https://www.slashroot.in/inode-and-its-structure-linux>
- Unix/Linux filesystem hierarchy image: https://www.linux.com/sites/lcom/files/styles/rendered_file/public/standard-unix-filesystem-hierarchy.png?itok=CVqmyk6P

- Networking illustration used in slide: <https://pbs.twimg.com/media/DuLLzQXcAAkFMV.jpg>

5. Practical Tasks

Task 1: Prepare Your Evidence Workspace

Create a structured workspace that will hold all commands, outputs, screenshots, and final evidence artifacts. Replace FIRSTNAME_LASTNAME with your real name.

```
cd ~
mkdir -p CIP-B101_Lab2A_FIRSTNAME_LASTNAME/{notes,evidence,scripts,outputs,screenshots,archive}
cd CIP-B101_Lab2A_FIRSTNAME_LASTNAME
pwd
ls -la
```

Evidence required: screenshot showing your working directory and the folders created.

Task 2: Inspect Linux Filesystem and OS Information

Use the commands below to observe Linux filesystem hierarchy, operating system information, and basic system structure.

```
clear
pwd
ls /
ls -la /
cat /etc/os-release
uname -a
whoami
id
```

Write a short explanation of the difference between /home, /etc, /proc, /sys, /dev, /bin and /tmp in your report.

Task 3: Understand Inodes and Disk Information

Create a test evidence file and inspect its inode and disk information. This shows that Linux tracks files using metadata structures, not only file names.

```
echo "This is Lab 2A evidence file created by FIRSTNAME LASTNAME" > evidence/lab2a_evidence.txt
ls -li evidence/lab2a_evidence.txt
stat evidence/lab2a_evidence.txt
df -h
df -i
```

In your report, explain what an inode is and why inode information may matter in digital forensics.

Task 4: Absolute and Relative Paths

Demonstrate that you understand absolute and relative paths by navigating between folders without relying only on the graphical interface.

```
cd ~/CIP-B101_Lab2A_FIRSTNAME_LASTNAME/evidence
pwd
cd ..
pwd
cd ./notes
pwd
cd ../evidence
pwd
# Display the absolute path of the evidence file
readlink -f lab2a_evidence.txt
```

In your report, write one absolute path and one relative path to the same evidence file.

Task 5: Create, View, Copy, Rename and Remove Files

Perform basic file management operations that forensic analysts use when organizing working copies and extracted evidence.

```
cd ~/CIP-B101_Lab2A_FIRSTNAME_LASTNAME

echo "Case Note 1: Initial Linux command line practice" > notes/case_note1.txt
echo "Case Note 2: Evidence handling requires careful documentation" > notes/case_note2.txt
cat notes/case_note1.txt
cat notes/case_note2.txt

cp notes/case_note1.txt evidence/case_note1_copy.txt
cp -r notes evidence/notes_backup
mv evidence/case_note1_copy.txt evidence/case_note1_renamed.txt
ls -la evidence

# Create and remove a temporary file
echo "temporary data" > evidence/temp_delete_me.txt
ls -la evidence/temp_delete_me.txt
rm evidence/temp_delete_me.txt
ls -la evidence
```

Evidence required: screenshot showing the files before and after rename/delete operations.

Task 6: Search Evidence Content with grep

Create a small evidence log and search it using grep. This introduces content searching, a key forensic skill.

```
cat > evidence/activity_log.txt <<'EOF'
2026-06-19 09:10 user=aminu action=login status=success ip=192.168.56.10
2026-06-19 09:25 user=guest action=file_copy status=success file=report.docx
2026-06-19 10:05 user=admin action=login status=failed ip=192.168.56.15
2026-06-19 10:20 user=aminu action=delete status=success file=temp.txt
2026-06-19 10:40 user=admin action=password_change status=success
EOF

cat evidence/activity_log.txt
grep "login" evidence/activity_log.txt
grep -n "failed" evidence/activity_log.txt
grep -Ei "delete|password" evidence/activity_log.txt
grep -Ei "192\.168\.56\.[0-9]+" evidence/activity_log.txt
```

In your report, identify which command shows failed login evidence and which command shows suspicious account activity.

Task 7: Search File Names and Display Directory Tree

Use tree and find to understand the evidence folder layout and locate files by name pattern.

```
cd ~/CIP-B101_Lab2A_FIRSTNAME_LASTNAME
tree
find . -type f -name "*.txt"
find . -type f -iname "*log*"
find . -type d -name "evidence"
```

Evidence required: screenshot showing the tree output and at least one find command result.

Task 8: Check Network Information Safely

Check your own system network settings and test local or approved connectivity only. Do not scan external networks.

```
ip addr
ip route
ping -c 4 127.0.0.1
ping -c 4 localhost
ss -lnt
nmap localhost -p 22,80,443 2>/dev/null || echo "nmap unavailable or no services detected"
```

In your report, write your loopback address and explain why localhost testing is safer than scanning unknown systems.

Task 9: Create and Execute a Simple Forensic Script

Create a shell script that records basic system and evidence-folder information into a text report.

```
cd ~/CIP-B101_Lab2A_FIRSTNAME_LASTNAME/scripts
nano lab2a_sys_summary.sh
```

Paste the script below into nano, save it, and exit.

```
#!/bin/sh
# CIP-B101 Lab 2A system summary script
# Student: FIRSTNAME LASTNAME

OUT="$HOME/CIP-B101_Lab2A_FIRSTNAME_LASTNAME/outputs/system_summary.txt"
{
    echo "==== CIP-B101 Lab 2A System Summary ====="
    echo "Date/Time: $(date)"
    echo "User: $(whoami)"
    echo "Hostname: $(hostname)"
    echo "Kernel: $(uname -a)"
    echo "Current Folder: $(pwd)"
    echo "==== Disk Space ====="
    df -h
    echo "==== Inode Usage ====="
    df -i
    echo "==== Evidence Workspace Tree ====="
    tree "$HOME/CIP-B101_Lab2A_FIRSTNAME_LASTNAME"
} > "$OUT"

echo "Summary saved to $OUT"

chmod +x lab2a_sys_summary.sh
ls -l lab2a_sys_summary.sh
./lab2a_sys_summary.sh
cat ../outputs/system_summary.txt | head -n 30
```

Evidence required: screenshot showing script permission, execution, and output file.

Task 10: Make the Script Callable from Another Folder Using PATH

Temporarily add the script folder to PATH and run the script from a different directory.

```
export PATH=$HOME/CIP-B101_Lab2A_FIRSTNAME_LASTNAME/scripts:$PATH
cd /tmp
```

```
lab2a_sys_summary.sh  
cat ~/CIP-B101_Lab2A_FIRSTNAME_LASTNAME/outputs/system_summary.txt | head -n 15
```

In your report, explain what PATH does and why Linux commands such as ls can run from different folders.

Task 11: Compress the Evidence Workspace

Compress your lab workspace for submission. You may use zip or tar.gz. Submit the compressed file along with your PDF report.

```
cd ~  
zip -r CIP-B101_Lab2A_FIRSTNAME_LASTNAME.zip CIP-B101_Lab2A_FIRSTNAME_LASTNAME  
  
tar -czvf CIP-B101_Lab2A_FIRSTNAME_LASTNAME.tar.gz CIP-B101_Lab2A_FIRSTNAME_LASTNAME  
  
ls -lh CIP-B101_Lab2A_FIRSTNAME_LASTNAME.zip CIP-B101_Lab2A_FIRSTNAME_LASTNAME.tar.gz
```

Evidence required: screenshot showing the compressed archive file and its size.

6. Screenshots Required

- ☐ Workspace creation showing folder structure.
- ☐ Linux OS and user information commands.
- ☐ Inode and stat output for the evidence file.
- ☐ Absolute/relative path demonstration.
- ☐ File creation, copy, rename, and delete operations.
- ☐ grep results showing login, failed, delete/password, and IP evidence.
- ☐ tree/find output showing file discovery.
- ☐ Network commands using localhost only.
- ☐ Shell script creation, permission change, execution, and output.
- ☐ PATH test showing script execution from a different folder.
- ☐ Final compressed archive file.

7. Student Report Template

Your report must be submitted as a PDF using the structure below.

Section	What to Include
Cover Page	Student name, course code, course title, lab title, date, instructor.
Introduction	Briefly explain the purpose of Linux command-line skills in digital forensics.
Lab Environment	Linux distribution, username, VM/host details, tools used.
Tasks Performed	Step-by-step explanation of each task and command used.
Evidence Screenshots	All required screenshots with captions.
Forensic Interpretation	Explain inodes, PATH, grep evidence search, and localhost network testing.
Challenges	Any issue encountered and how you solved it.
Conclusion	Skills gained and how they support later forensic labs.

8. Questions to Answer in the Report

1. What is a Linux inode and why is it important?
2. What is the difference between an absolute path and a relative path?
3. Why does Linux say “everything is a file”? Give two examples from /dev, /proc, /sys, or /etc.
4. What is the purpose of the PATH variable?
5. Which grep command did you use to identify failed login evidence?
6. Why should a forensic trainee avoid scanning unauthorized external systems?
7. What is the difference between zip and tar.gz in this lab context?

9. Submission Checklist

- ☐ PDF lab report named CIP-B101_Lab2A_Report_FIRSTNAME_LASTNAME.pdf.
- ☐ Compressed evidence workspace named CIP-B101_Lab2A_FIRSTNAME_LASTNAME.zip or .tar.gz.
- ☐ All screenshots included and labelled in the report.
- ☐ All commands explained in your own words.
- ☐ No unauthorized scanning or external testing performed.

10. Marking Rubric

Assessment Area	Marks	Criteria
Workspace setup and organization	10	Correct folder structure, clean naming, organized outputs.
Linux filesystem and inode understanding	15	Correct use of ls, stat, df, inode explanation, filesystem interpretation.
Path and file management tasks	15	Correct absolute/relative path use, file creation, copy, rename, deletion.
Evidence search using grep/find/tree	20	Correct search commands, meaningful evidence identification, good screenshots.
Network and system information	10	Safe localhost testing, correct interpretation of IP/route/listening ports.
Shell scripting and PATH task	15	Functional script, executable permission, PATH execution from another folder.
Report quality and screenshots	10	Clear explanations, labelled screenshots, professional formatting.
Evidence archive submission	5	Correct compressed workspace submitted.
Total	100	Full practical completion.

Source Material: This lab was prepared from the “Linux OS For Forensics / Linux Command Line Tutorial” slide material for CIP-B101 Basic Computer Skills for Digital Forensics.